

National University of Computer & Emerging Sciences

Chiniot – Faisalabad Campus

CL3001 – Computer Networks

Lab Report

Lab 2 & Week 2

Course	CL3001 – Computer Networks
Lab / Week	Lab 2 – Week 2
Instructor	Esha Iftikhar
Topic	Linux CLI Utilities
Tasks	Task 1: Incident Response Task 2: Forensic Investigation

■ Task 1 – Incident Response Investigation

You are a Junior System Administrator for a small tech company. The lead developer reports that a critical application crashed last night around **02:00 AM**. They suspect a specific error occurred, but the logs are messy and some files seem to have been tampered with or moved. Your goal is to navigate the server, locate the relevant log files, extract the specific error messages, and fix a permissions issue that is preventing the application from restarting.

Objectives

- Navigate the Linux file system (**cd**, **ls**, **pwd**).
- Search for files based on criteria (**find**).
- Search within files for specific text patterns (**grep**).
- Sort and filter data streams (**sort**, **uniq**, **awk**).
- Manage file permissions and ownership (**chmod**, **chown**).
- Archive and compress data (**tar**).

■■ **Note:** Run the setup script below in your terminal before starting the steps.

```
mkdir -p ~/lab_incident/logs/archive
mkdir -p ~/lab_incident/conf
mkdir -p ~/lab_incident/data

for i in {1..5}; do
echo "2023-10-2$i 01:00:00 INFO Service started" >
~/lab_incident/logs/server_$i.log
echo "2023-10-2$i 02:15:00 ERROR ConnectionRefused: DB_HOST_1" >>
~/lab_incident/logs/server_$i.log
echo "2023-10-2$i 03:00:00 INFO Service stopped" >>
~/lab_incident/logs/server_$i.log
done

echo "SECRET_KEY=12345" > ~/lab_incident/conf/.hidden_config
echo "#!/bin/bash" > ~/lab_incident/data/restart_service.sh
echo "echo 'Service Restarted'" >> ~/lab_incident/data/restart_service.sh
chmod 000 ~/lab_incident/data/restart_service.sh

echo -e "User:Alice IP:192.168.1.5\nUser:Bob
IP:192.168.1.10\n...\nUser:Charlie IP:192.168.1.20" >
~/lab_incident/logs/access.raw
```

Step 1: Reconnaissance

Navigate to **~/lab_incident** and list all files in **conf**, including hidden files.

```
cd ~/lab_incident
ls -la conf/
```

Question: What is the name of the hidden file inside conf?

```
wahab@wahab-HP-EliteBook-840-G3:~/cn_lab_2$ pwd
/home/wahab/cn_lab_2
ls
cd ~/lab_incident
ls
task1.txt Task2.txt
conf 'conf'$'\r' data 'data'$'\r' logs
wahab@wahab-HP-EliteBook-840-G3:~/lab_incident$ cd logs
ls
cd archive
pwd
access.raw 'archive'$'\r' server_2.log server_4.log
archive server_1.log server_3.log server_5.log
/home/wahab/lab_incident/logs/archive
```

Figure 1.1 – Directory navigation & listing

```
wahab@wahab-HP-EliteBook-840-G3:~/lab_incident/logs/archive$ find ~/lab_incident -name "*.log"
/home/wahab/lab_incident/logs/server_5.log
/home/wahab/lab_incident/logs/server_4.log
/home/wahab/lab_incident/logs/server_3.log
/home/wahab/lab_incident/logs/server_1.log
/home/wahab/lab_incident/logs/server_2.log
wahab@wahab-HP-EliteBook-840-G3:~/lab_incident/logs/archive$ find ~/lab_incident -type f
/home/wahab/lab_incident/conf/.hidden_config
/home/wahab/lab_incident/logs/access.raw
/home/wahab/lab_incident/logs/server_5.log
/home/wahab/lab_incident/logs/server_4.log
/home/wahab/lab_incident/logs/server_3.log
/home/wahab/lab_incident/logs/server_1.log
/home/wahab/lab_incident/logs/server_2.log
/home/wahab/lab_incident/data/restart_service.sh
wahab@wahab-HP-EliteBook-840-G3:~/lab_incident/logs/archive$ find ~/lab_incident -name "*"
/home/wahab/lab_incident/conf/.hidden_config
```

Figure 1.2 – Hidden file revealed in conf/

Step 2: Locating the Errors

Navigate to `~/lab_incident/logs` and search all `.log` files for the string **ERROR**. Save the results to `error_report.txt`.

```
cd ~/lab_incident/logs
grep 'ERROR' *.log > error_report.txt
cat error_report.txt
```

```
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ grep "ERROR" ~/lab_
incident/logs/*.log
/home/wahab/lab_incident/logs/server_1.log:2023-10-21 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_2.log:2023-10-22 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_3.log:2023-10-23 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_4.log:2023-10-24 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_5.log:2023-10-25 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ grep "02:15" ~/lab_
incident/logs/*.log
/home/wahab/lab_incident/logs/server_1.log:2023-10-21 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_2.log:2023-10-22 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_3.log:2023-10-23 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_4.log:2023-10-24 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_5.log:2023-10-25 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
```

Figure 2.1 – grep ERROR across all log files

```
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ grep -i error ~/lab_
incident/logs/*.log
/home/wahab/lab_incident/logs/server_1.log:2023-10-21 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_2.log:2023-10-22 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_3.log:2023-10-23 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_4.log:2023-10-24 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
/home/wahab/lab_incident/logs/server_5.log:2023-10-25 02:15:00 ERROR ConnectionR
efused: DB_HOST_1
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ grep "ERROR" ~/lab_
incident/logs/*.log | awk '{print $6}'
5
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ grep "ERROR" ~/lab_
incident/logs/*.log | awk '{print $6}' | sort | uniq -c
5
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ cat ~/lab_incident/
logs/access.raw
```

Figure 2.2 – error_report.txt contents

Step 3: Data Cleaning

The **access.raw** file contains duplicate user-access entries. Display unique lines and extract IP addresses.

```
cat access.raw
sort access.raw | uniq
sort access.raw | uniq | awk '{print $2}' | cut -d: -f2 > unique_ips.txt
cat unique_ips.txt
```

```

5
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ cat ~/lab_incident/
logs/access.raw
User:Alice IP:192.168.1.5 refused: 10_HOST_1
User:Bob IP:192.168.1.10
User:Alice IP:192.168.1.5
User:Charlie IP:192.168.1.20 refused: 10_HOST_1
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ awk '{print $2}' ~/
lab_incident/logs/access.raw | sort | uniq -c
      1 IP:192.168.1.10
      1 IP:192.168.1.20
      2 IP:192.168.1.5

```

Figure 3.1 – Unique entries from access.raw

```

wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ ~/lab_incident/data
/restart_service.sh
bash: /home/wahab/lab_incident/data/restart_service.sh: Permission denied
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ ls -l ~/lab_inciden
t/data/
total 4
----- 1 wahab wahab 37 Jan 29 09:51 restart_service.sh
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ chmod +x ~/lab_inci
dent/data/restart_service.sh
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ ~/lab_incident/data
/restart_service.sh
/bin/bash: /home/wahab/lab_incident/data/restart_service.sh: Permission denied
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ chmod 755 ~/lab_inc
ident/data/restart_service.sh
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ ls -l ~/lab_inciden
t/data/restart_service.sh
-rwxr-xr-x 1 wahab wahab 37 Jan 29 09:51 /home/wahab/lab_incident/data/restart_s
ervice.sh
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ ~/lab_incident/data
/restart_service.sh
Service Restarted

```

Figure 3.2 – Extracted unique IPs saved to unique_ips.txt

Step 4: Fixing the Script Permissions

Navigate to `~/lab_incident/data`. The restart script is locked. Fix its permissions to **700** and execute it.

```

cd ~/lab_incident/data
cat restart_service.sh # Should say: Permission denied
chmod 700 restart_service.sh
ls -l restart_service.sh
./restart_service.sh

```

```

Service Restarted
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident/logs/archive$ cd ~/lab_incident
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident$ tar -czvf logs_backup.tar.gz log
s/
logs/
logs/access.raw
logs/archive/
logs/server_5.log
logs/server_4.log
logs/server_3.log
logs/archive\r/
logs/server_1.log
logs/server_2.log
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident$ tar -tzvf logs_backup.tar.gz
drwxrwxr-x wahab/wahab    0 2026-01-29 09:51 logs/
-rw-rw-r-- wahab/wahab   106 2026-01-29 09:51 logs/access.raw
drwxrwxr-x wahab/wahab    0 2026-01-29 09:51 logs/archive/
-rw-rw-r-- wahab/wahab   137 2026-01-29 09:51 logs/server_5.log
-rw-rw-r-- wahab/wahab   137 2026-01-29 09:51 logs/server_4.log
-rw-rw-r-- wahab/wahab   137 2026-01-29 09:51 logs/server_3.log
drwxrwxr-x wahab/wahab    0 2026-01-29 09:43 logs/archive\r/
-rw-rw-r-- wahab/wahab   137 2026-01-29 09:51 logs/server_1.log
-rw-rw-r-- wahab/wahab   137 2026-01-29 09:51 logs/server_2.log
wahab@wahab-HP-EliteBook-840-G3: ~/lab_incident$

```

Figure 4.1 – Permission denied, then chmod 700 applied

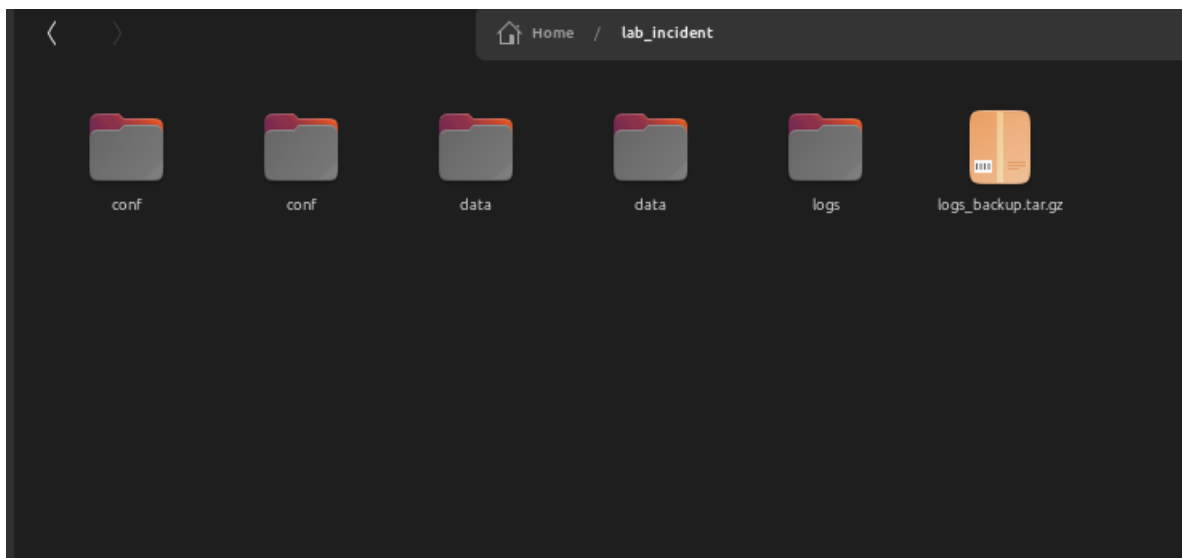


Figure 4.2 – Script executes successfully

Step 5: Archiving Evidence

From `~/lab_incident`, create a compressed tarball named **incident_report.tar.gz** containing the logs directory and unique_ips.txt.

```

cd ~/lab_incident
tar -czvf incident_report.tar.gz logs/ logs/unique_ips.txt
tar -tzf incident_report.tar.gz # verify contents

```

```
wahab@wahab-HP-EliteBook-840-G3: /lab_incident/data$ cd ~/lab/logs
wahab@wahab-HP-EliteBook-840-G3: ~/lab/logs$ find . -type f -mtime -1 -ls
 423291    4 -rw-rw-r--  1 wahab  wahab    131 Jan 29 10:25 ./access.raw
 423282    4 -rw-rw-r--  1 wahab  wahab    245 Jan 29 10:25 ./server_5.log
 423281    4 -rw-rw-r--  1 wahab  wahab    244 Jan 29 10:25 ./server_4.log
 423280    4 -rw-rw-r--  1 wahab  wahab    244 Jan 29 10:25 ./server_3.log
 423278    4 -rw-rw-r--  1 wahab  wahab    244 Jan 29 10:25 ./server_1.log
 423279    4 -rw-rw-r--  1 wahab  wahab    244 Jan 29 10:25 ./server_2.log
wahab@wahab-HP-EliteBook-840-G3: ~/lab/logs$ ls -lt
total 28
-rw-rw-r--  1 wahab wahab  131 Jan 29 10:25 access.raw
-rw-rw-r--  1 wahab wahab  245 Jan 29 10:25 server_5.log
-rw-rw-r--  1 wahab wahab  244 Jan 29 10:25 server_1.log
-rw-rw-r--  1 wahab wahab  244 Jan 29 10:25 server_2.log
-rw-rw-r--  1 wahab wahab  244 Jan 29 10:25 server_3.log
-rw-rw-r--  1 wahab wahab  244 Jan 29 10:25 server_4.log
drwxrwxr-x  2 wahab wahab 4096 Jan 29 10:25 archive
```

Figure 5.1 – Creating and verifying incident_report.tar.gz

■ Task 2 – Forensic Investigation

The company recently detected unusual activity on the server. Multiple users report intermittent failures in the application, and some log files have been modified. Your task is to perform a forensic investigation: identify suspicious activity, generate a report of affected users, and secure the system by restoring correct permissions and compressing the findings.

Objectives

- Advanced file searching and pattern matching (**find, grep, egrep**).
- Data extraction and reporting (**awk, sed, sort, uniq**).
- Permission and ownership management (**chmod, chown**).
- Log file analysis and filtering.
- Archiving and compressing multiple files.
- Optional: Detect recently modified files (**ls -lt, stat**).

■■ **Note:** Run the Task 2 setup script in your terminal before proceeding.

```
mkdir -p ~/lab/logs/archive
mkdir -p ~/lab/conf
mkdir -p ~/lab/data

for i in {1..5}; do
LOG_FILE=~/.lab/logs/server_${i}.log
echo "2026-01-2${i} 00:50:00 INFO Pre-check completed" > $LOG_FILE
echo "2026-01-2${i} 01:30:00 WARNING Disk usage high" >> $LOG_FILE
echo "2026-01-2${i} 02:00:00 FAILED LOGIN User:Alice IP:192.168.1.${(i*2)}" >>
$LOG_FILE
echo "2026-01-2${i} 02:15:00 ERROR ConnectionRefused: DB_HOST_${i}" >> $LOG_FILE
echo "2026-01-2${i} 03:00:00 INFO Service stopped" >> $LOG_FILE
done

echo "DB_HOST=DB_HOST_1\nDB_PORT=3306\nAPP_MODE=production" >
~/lab/conf/.config_backup
echo "DB_HOST=DB_HOST_1\nDB_PORT=3306\nAPP_MODE=development" >
~/lab/conf/app.conf
echo "#!/bin/bash" > ~/lab/data/restart_service.sh
chmod 000 ~/lab/data/restart_service.sh
echo "#!/bin/bash" > ~/lab/data/backup.sh
chmod 750 ~/lab/data/backup.sh
```

Step 1: Identify Recently Modified Files

Navigate to `~/lab/logs`. List files modified in the last 24 hours and identify those modified between 01:00–05:00 AM.

```
cd ~/lab/logs
find . -mtime -1 -name '*.log' | xargs ls -lt
ls -lt --time-style='+%H:%M' *.log
```

Question: Which file(s) were modified in the 01:00 AM – 05:00 AM time window?

```
wahab@wahab-HP-EliteBook-840-G3: /lab/log$ grep -E "WARNING|FAILED LOGIN" ~/lab/logs/*.log > ~/lab/suspicious_activity.txt
wahab@wahab-HP-EliteBook-840-G3: /lab/log$ cat ~/lab/suspicious_activity.txt
/home/wahab/lab/logs/server_1.log:2026-01-21 01:30:00 WARNING Disk usage high
/home/wahab/lab/logs/server_1.log:2026-01-21 02:00:00 FAILED LOGIN User:Alice IP:192.168.1.2
/home/wahab/lab/logs/server_2.log:2026-01-22 01:30:00 WARNING Disk usage high
/home/wahab/lab/logs/server_2.log:2026-01-22 02:00:00 FAILED LOGIN User:Alice IP:192.168.1.4
/home/wahab/lab/logs/server_3.log:2026-01-23 01:30:00 WARNING Disk usage high
/home/wahab/lab/logs/server_3.log:2026-01-23 02:00:00 FAILED LOGIN User:Alice IP:192.168.1.6
/home/wahab/lab/logs/server_4.log:2026-01-24 01:30:00 WARNING Disk usage high
/home/wahab/lab/logs/server_4.log:2026-01-24 02:00:00 FAILED LOGIN User:Alice IP:192.168.1.8
/home/wahab/lab/logs/server_5.log:2026-01-25 01:30:00 WARNING Disk usage high
/home/wahab/lab/logs/server_5.log:2026-01-25 02:00:00 FAILED LOGIN User:Alice IP:192.168.1.10
wahab@wahab-HP-EliteBook-840-G3: /lab/log$ awk '{print $5, $7}' ~/lab/suspicious_activity.txt
usage
User:Alice
usage
User:Alice
usage
User:Alice
usage
User:Alice
usage
User:Alice
usage
User:Alice
# Extract username + IP (adjust field numbers if needed)
wahab@wahab-HP-EliteBook-840-G3: /lab/log$ awk '{print $5, $7}' ~/lab/suspicious_activity.txt | sort | uniq > ~/lab/affected_users.txt
```

Figure T2-1.1 – Recently modified log files

```
wahab@wahab-HP-EliteBook-840-G3: /lab/data$ cd -/lab/data
wahab@wahab-HP-EliteBook-840-G3: /lab/data$ ls -l
total 8
-rwxr-xr-x 1 wahab wahab 36 Jan 29 18:25 backup.sh
-rwxr-xr-x 1 wahab wahab 37 Jan 29 18:25 restart_service.sh
wahab@wahab-HP-EliteBook-840-G3: /lab/data$ chmod 700 restart_service.sh
chmod 754 backup.sh
wahab@wahab-HP-EliteBook-840-G3: /lab/data$ ./restart_service.sh
Service Restarted
```

Figure T2-1.2 – Files in the suspicious time window

Step 2: Extract Suspicious Activity

Search all .log files for **WARNING** or **FAILED LOGIN** lines, extract usernames and IPs, remove duplicates.

```
egrep 'WARNING|FAILED LOGIN' *.log > suspicious_activity.txt
```

```
cat suspicious_activity.txt
```

```
awk '/FAILED LOGIN/{print $5, $6}' suspicious_activity.txt | sort | uniq >
affected_users.txt
```

```
cat affected_users.txt
```

```
wahab@wahab-HP-EliteBook-840-G3: ~/lab/conf$ cd ~/lab/conf
wahab@wahab-HP-EliteBook-840-G3: ~/lab/conf$ ls -la
total 16
drwxrwxr-x 2 wahab wahab 4096 Jan 29 10:25
drwxrwxr-x 5 wahab wahab 4096 Jan 29 10:37
-rw-rw-r-- 1 wahab wahab 52 Jan 29 10:25 app.conf
-rw-rw-r-- 1 wahab wahab 51 Jan 29 10:25 .config_backup
wahab@wahab-HP-EliteBook-840-G3: ~/lab/conf$ diff app.conf .config_backup
3c3
< APP_MODE=development
...
> APP_MODE=production
wahab@wahab-HP-EliteBook-840-G3: ~/lab/conf$ cp .config_backup app.conf
wahab@wahab-HP-EliteBook-840-G3: ~/lab/conf$ cat app.conf
DB_HOST=DB_HOST_1
DB_PORT=3306
APP_MODE=production
```

Figure T2-2.1 – suspicious activity.txt generated

```

wahab@wahab-HP-EliteBook-B40-G3: ~ $ cd ~/lab
wahab@wahab-HP-EliteBook-B40-G3: ~ $ ls
affected_users.txt  data  suspicious_activity.txt
wahab@wahab-HP-EliteBook-B40-G3: ~ $ tar -czvf forensic_report.tar.gz \
suspicious_activity.txt \
affected_users.txt \
logs \
conf \
suspicious_activity.txt
affected_users.txt
logs/
logs/access.raw
logs/archive/
logs/server_5.log
logs/server_4.log
logs/server_3.log
logs/server_1.log
logs/server_2.log
conf/
conf/.config_backup
conf/app.conf
wahab@wahab-HP-EliteBook-B40-G3: ~ $ tar -tzvf forensic_report.tar.gz
suspicious_activity.txt
affected_users.txt
logs/
logs/access.raw
logs/archive/
logs/server_5.log
logs/server_4.log
logs/server_3.log
logs/server_1.log
logs/server_2.log
conf/
conf/.config_backup
conf/app.conf
wahab@wahab-HP-EliteBook-B40-G3: ~ $

```

Figure T2-2.2 – affected_users.txt (deduplicated)

Step 3: Correct File Permissions

Fix permissions in ~/lab/data: **restart_service.sh** → **700**, **backup.sh** → **754**. Test execution.

```
cd ~/lab/data
```

```
ls -l
```

```
chmod 700 restart_service.sh
```

```
chmod 754 backup.sh
```

```
ls -l
```

```
./restart_service.sh
```

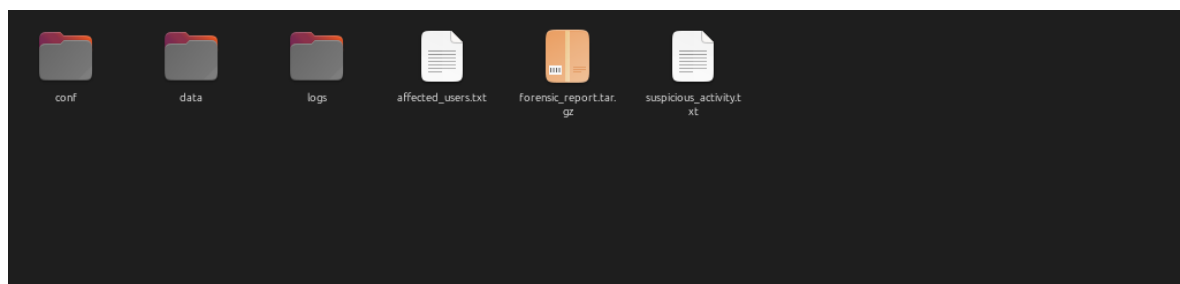


Figure T2-3 – Permissions corrected and script tested

Step 4: Replace Tampered Config Files

Compare **app.conf** with hidden backup **.config_backup**. If different, restore from backup.

```
cd ~/lab/conf
```

```
diff app.conf .config_backup
```

```
cp .config_backup app.conf
```

```
cat app.conf # verify APP_MODE=production
```

The **diff** command will reveal that **app.conf** has **APP_MODE=development** while the backup correctly has **APP_MODE=production**. Restoring from backup fixes the tampered file.

Step 5: Package Evidence

From ~/lab, create **forensic_report.tar.gz** containing all evidence files and directories.

```
cd ~/lab
```

```
tar -czvf forensic_report.tar.gz suspicious_activity.txt affected_users.txt  
logs/ conf/
```

```
tar -tzf forensic_report.tar.gz # verify
```